

Experiment 8

8.1.1 Multiplication of Two Square Matrix

Algorithm :

Step 1: Start

Step 2: Input Dimension (n)

Step 3: A=[]

 Create an empty list A

 Set loop a in range 0 to n:

 Input a row of integers and append to A

Step 4: B=[]

 Create an empty list B

 Set loop b in range 0 to n:

 Input a row of integers and append to B

Step 5: Result =[]

 Create an empty list result

 Set loop r in range 0 to n:

 Append a list of n zeros to result

Step 6: Multiplication of Matrix

 Set loop i in range 0 to n:

 Set loop j in range 0 to n :

 Set loop k in range 0 to n :

 result[i][j] += (A[i][k] * B[k][j])

Step 7: Print Resultant Matrix

 Set loop i in range \$0\$ to \$n\$:

 Set loop j in range \$0\$ to \$n\$:

 If j is the last index :

 Print result[i][j] (end="")

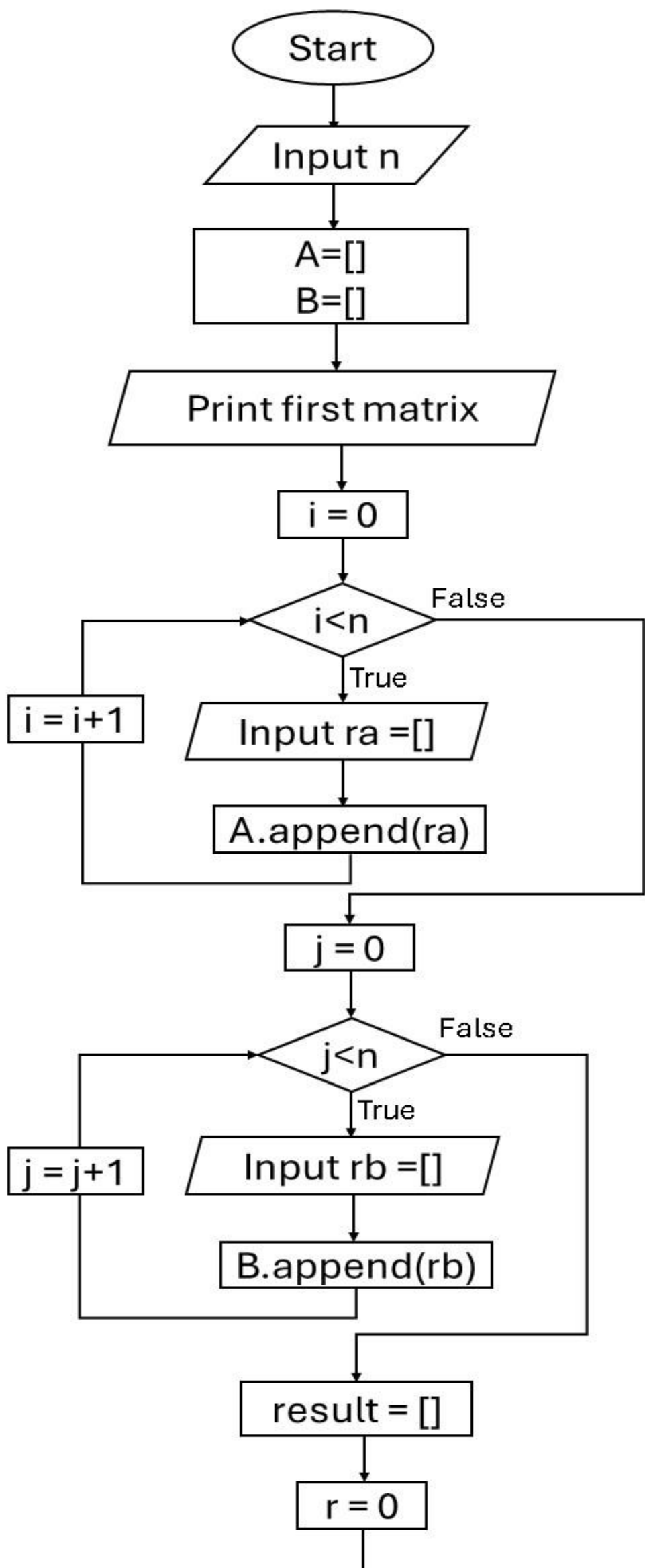
 Else:

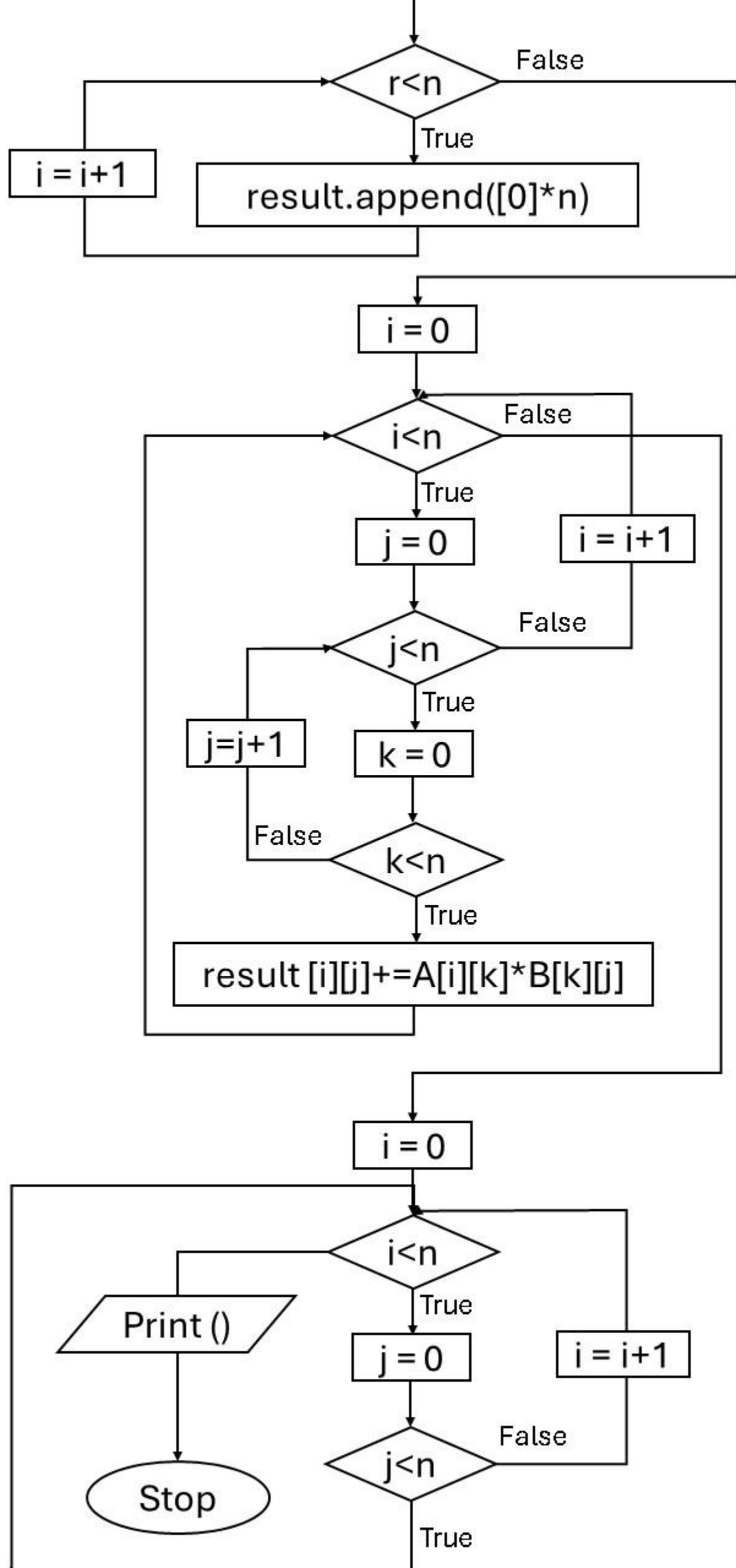
 Print result[i][j](end=" ")

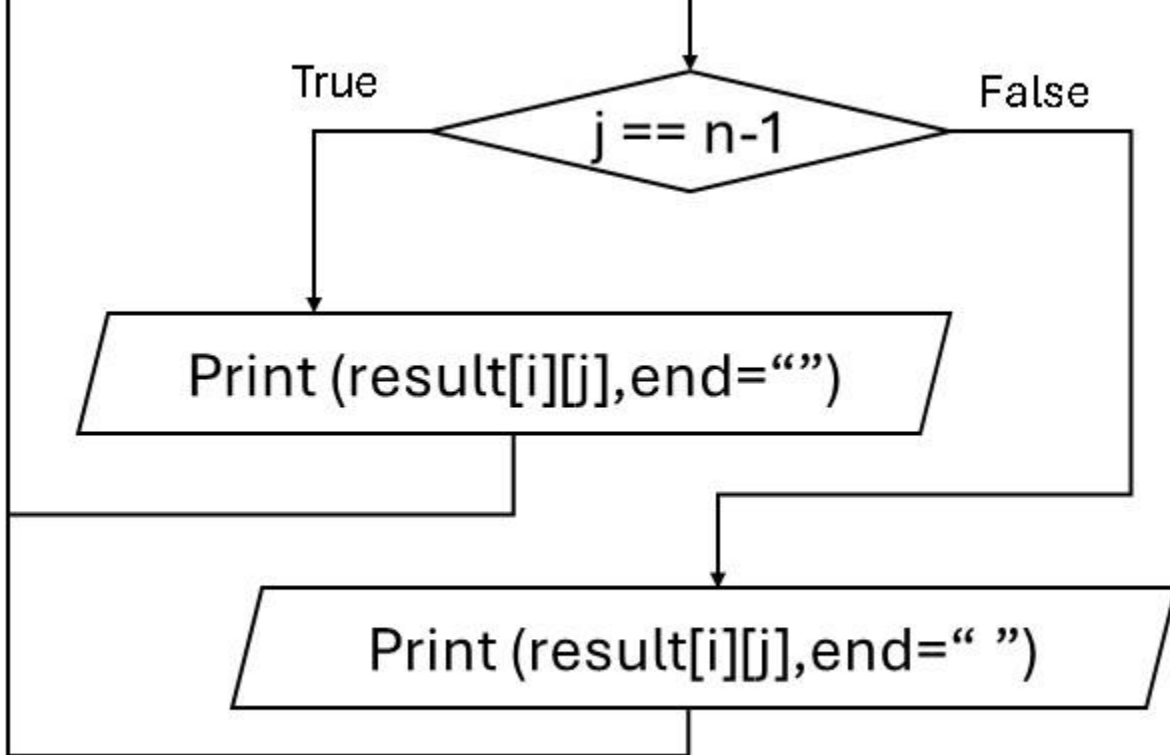
 Print a new line.

Step 8: Stop

Flowchart :







Execution :

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8.1.1. Multiplication of Two Square Matrices

16:37

Write a Python program to perform the multiplication of two square matrices. The program should take two matrices as input and output their product.

Input Format:

- The first line prints the "dimension: " and reads an integer n , representing the dimensions of the square matrices ($n \times n$).
- The second line prints: "first matrix:" and then reads n lines, each containing n integers separated by spaces, representing the rows of the first matrix.
- Next line prints: "second matrix:" and then reads n lines, each containing n integers separated by spaces, representing the rows of the second matrix.

Output Format:

- Print "Resultant Matrix:" followed by n lines, each containing n integers separated by spaces, representing the product of the two matrices.

Sample Test Cases

Multiplicat...

Submit

```

1 n=int(input("dimension: "))
2 A=[]
3 print("first matrix:")
4 for a in range(n):
5     ra=list(map(int,input().split()))
6     A.append(ra)
7 B=[]
8 print("second matrix:")
9 for b in range(n):
10    rb=list(map(int,input().split()))
11    B.append(rb)
12

```

Average time	Maximum time	Test Results
0.015 s	0.016 s	2 out of 2 shown test case(s) passed
14.50 ms	16.00 ms	2 out of 2 hidden test case(s) passed

Test case 1 15 ms

Test case 2 13 ms

Terminal

Test cases

< Prev

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```
n=int(input("dimension: "))
A=[]
print("first matrix:")
for a in range(n):
    ra=list(map(int,input().split()))
    A.append(ra)
B=[]
print("second matrix:")
for b in range(n):
    rb=list(map(int,input().split()))
    B.append(rb)

result=[]
for r in range(n):
    result.append([0]*n)

for i in range(n):
    for j in range(n):
        for k in range(n):
            result[i][j]+=A[i][k]*B[k][j]

print("Resultant Matrix:")
for i in range(n):
    for j in range(n):
        if j==n-1:
            print(result[i][j],end="")
        else:
            print(result[i][j],end=" ")
    print()
```